2400030075 HOME ASSIGNMENT-II. Ch. Shirdi Sai. (1)

1. Use Branch and Bound Method to Maximize $Z = 5x_1 + 3x_2$ Subject to: $4x_1 + 5x_2 <= 6$, $x_1 + 3x_2 <= 8$, $7x_2 <= 8$. where x_1, x_2 are non negative integers?

Ans:

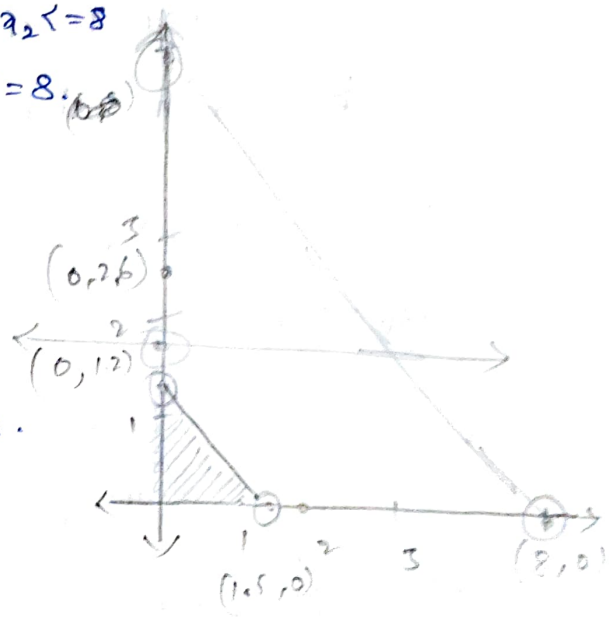
Given Maximize $Z = 5x_1 + 3x_2$.

Equations are $4x_1 + 5x_2 <= 6$

$x_1 + 3x_2 <= 8$

$7x_2 <= 8$.

$x_2 = 8/7$. $x_1 = 0$.



$4x_1 + 5x_2 = 6$.

put $x_1 = 0$ $5x_2 = 6$

$x_2 = 6/5 = 1.2$.

$x_2 = 0$ $x_1 = 3/2 = 1.5$

$x_1 + 3x_2 <= 8$.

$x_1 = 0$ $3x_2 = 8$

$x_2 = 8/3 = 2.6$.

$x_2 = 0$ $x_1 = 8$.

feasible point.

x_1	x_2	Z.
0	0.	0
0	1.2	3.6
1.5	0	7.5

the Solution is $(1.5, 0)$

Consider $x_1 \geq 1$ and $x_2 \leq 2$

and solve the Graphical method we get.

$x_1 = 1$, $x_2 = 0$.

2. Use Newton's method to find the minimum value of $f(x) = x^2 - 6x + 20$ with intial guess $x_0 = 0$?

Ans:

$f(x) = x^2 - 6x + 20$.

$x_0 = 0$.

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}.$$

$$f'(x) = \frac{d}{dx}(x^2 - 6x + 20)$$

$f'(x) = 2x - 6$.

$$f''(x) = \frac{d}{dx}(2x - 6)$$

$f''(x) = 2$.

iteration 1:

$x_0 = 0$.

$f'(0) = 2(0) - 6 = -6$.

$f''(0) = 2$.

3. Solve the objective function by Gomery cut plane method. objective function MAX $Z = x_1 + 4x_2$, Subject to $6x_1 + 4x_2 <= 24$; $2x_1 + 2x_2 <= 6$, where $x_1, x_2 \geq 0$ and non negative integers?

Ans:

$$Z = x_1 + 4x_2$$

$$6x_1 + 4x_2 <= 24$$

$$2x_1 + 2x_2 \leq 6$$

$$x_1, x_2 \geq 0.$$

Slack variables.

$$6x_1 + 4x_2 + S_1 = 24$$

$$2x_1 + 2x_2 + S_2 = 6.$$

$$Z = x_1 + 4x_2$$

Simplex Table.

	B.V	1	4	0	0		
		x_1	x_2	S_1	S_2	Sol	Ratio
0	S_1	6	4	1	0	24	$24/4 = 6$
0	S_2	2	2	0	1	6	$6/2 = 3$
	Z_j	0	0	0	0		
	$C_j - Z_j$	1	4	0	0		$C_j - Z_j \not\leq 0$

Apply Newton's formula.

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x_1 = 0 - \frac{-6}{2}.$$

$$x_1 = 3.$$

iteration 2:

$$x_1 = 3$$

$$f'(3) = 2(3) - 6 = 0$$

$$f''(3) = 2$$

$$x_2 = 3 - \frac{0}{2}$$

$$x_2 = 3.$$

Since $x_2 - x_1$

Converged. Minimum Value. $x = 3$.

$$f(3) = 3^2 - 6(3) + 20$$

$$= 9 - 18 + 20.$$

$$f(3) = 11$$

Minimum Value $f(x) = 11$ at $x = 3$.

Entering variable x_2

leaving variable S_2.

	B.V	1	4	0	0	
		x_1	x_2	S_1	S_2	Sol.
0	S_1	2	0	1	-2	12.
4	x_2	1	1	0	1/2	3.
-	z_j	4	4	0	2	
	$c_j - z_j$	-3	0	0	-2.	

$c_j - z_j \leq 0$ ✓

$\left.\begin{array}{l} x_2 = 3 \\ x_1 = 0. \end{array}\right\}$ optimal solution.

Maximum value $z = x_1 + 4x_2 = 0 + 4(3) \Rightarrow 12$

$\boxed{z = 12}$

4. Algorithmic steps involved in newton's method for optimization with an example?

Ans: Newton's method is used to find minimum/maximum of a function.

Step 1: choose intial guess x_0.

Step 2: Compute first derivative $f'(x)$.

Step 3: Compute second derivative $f''(x)$.

Step 4: Apply Newtons formula.

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}.$$

Step 5: Check convergence.

if $|x_{k+1} - x_k| < \epsilon$

Example:

$$f(x) = x^2 - 4x + 6.$$

intial $x_0 = 0$

$$f'(x) = 2x - 4$$

$$f''(x) = 2.$$

iteration 1: $x_0 = 0$

$$f'(0) = -4$$

$$f''(0) = 2.$$

$$x_1 = 0 - \frac{-4}{2}.$$

$$x_1 = 2.$$

iteration 2:

$$x_1 = 2$$

$$f'(2) = 0$$

$$x_2 = 2$$

$\boxed{\text{at } x = 2}$

Minimum Value: $f(2) = 2^2 - 4(2) + 6$ $\quad f(2) = 2.$

$f(2) = 4 - 8 + 6$

5. Steps involved in solving an integer programming problems using the Branch and Bound method?

Ans:

Steps:-

1. Formulat the problem.
   - Write the objective function, Constraints and specify integer variables.
2. Solve as linear programming problem.
   - ignore integer condition and solve using Simplex method.
3. check the solution.
   - if the solution is integer → stop
     if solution is fractional → Go to next step.
4. Branching.
   - select the variable with fractional value and Create two new Contraints.
5. Solve Sub problems
   - Solve each new problem using Simplex method.
6. Bounding.
   Compare solutions:
   - Remove infeasible solutions.

- keep best integer solution.

7. Repeat.

- if solution still fractional, branch again.

8. Stop.

- when all solutions become integer, choose best value as optimal.